

Markscheme

Mathematics: analysis and approaches
Practice question bank

129. (a)

Given $z_1 = 2 + 3i$ and $z_2 = 1 - i$.

Find $\frac{z_1}{z_2}$ in the form $a + bi$.

multiply by the conjugate: $\frac{(2 + 3i)(1 + i)}{(1 - i)(1 + i)}$ (M1)

$$= \frac{-1 + 5i}{2} \quad \text{A1}$$

$$= -0.5 + 2.5i \quad \text{A1}$$

[4 marks]